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HOLDER FOR STABILIZING A LINEAR SHAPED CHARGE DURING EPOXY CURING

Abstract

Disclosed is a holder for stabilizing a linear shaped charge during epoxy curing. The device includes a support with a rectangular midsection having a first and a second end, and a central aperture; a pair of stair-step risers extending down from the first and the second end; and a pair of curved legs extending from the stair-step risers. The pair of curved legs lock the support onto a container. A linear shaped charge is positioned within the central aperture and extends downward into the container where it is stabilized while a curing agent within the container hardens. The support removes the need for constantly checking the alignment of the liner shaped charge while curing and prevents the need for sanding, both of which save considerable time when compared to previous methods. The support is designed with a raised midsection to prevent it from being bonded to the epoxy during the curing process, thereby allowing it to be reused.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application claims priority to U.S. Provisional Patent Application Ser. No. 63/535,143, filed Aug. 29, 2023, entitled “LINEAR SHAPED CHARGE SUPPORT FOR EPOXY CURING,” the disclosure of which is expressly incorporated by reference herein.

FIELD OF THE INVENTION
[0003] The field of invention relates generally to supports. More particularly, it pertains to linear shaped charge support for epoxy curing.
BACKGROUND
[0004] A linear shaped charge (LSC) is an explosive device with a continuous explosive core enclosed in a chevron or V-shaped metal lining. The lining causes tamping upon detonation, which focuses an explosive high pressure wave as it becomes incident to the side wall causes the metal liner to collapse, which creates cutting force via a continuous, knife-like jet.
[0005] Linear shaped charges are potted in a curing agent (epoxy) for quality control, lot acceptance testing, and analysis of its structure. Previous methods for potting require the LSCs to be sanded flat on one end by hand. As can be appreciated, sanding requires a significant amount of time due to the number of samples in a given lot. After sanding, the LSCs are placed upright in a pot with epoxy for curing. Due to the shape of the LSCs, they are prone to falling over while curing. As a result, the LSCs need to be frequently monitored, and their position corrected before the epoxy cures. The process of sanding and potting/curing takes approximately 48 hours. As can be appreciated from the above, a device that can support a linear shaped charge during potting is desirable.
SUMMARY OF THE INVENTION
[0006] Disclosed is a holder for stabilizing a linear shaped charge during epoxy curing. The device includes a support with a rectangular midsection having a first and a second end, and a central aperture; a pair of stair-step risers extending down from the first and the second end; and a pair of curved legs extending from the stair-step risers. The pair of curved legs lock the support onto a container. A linear shaped charge is positioned within the central aperture and extends downward into the container where it is stabilized while a curing agent within the container hardens. The support removes the need for constantly checking the alignment of the liner shaped charge while curing and prevents the need for sanding, both of which save considerable time when compared to previous methods. The support is designed with a raised midsection to prevent it from being bonded to the epoxy during the curing process, thereby allowing it to be reused.
[0007] Additional features and advantages of the present invention will become apparent to those skilled in the art upon consideration of the following detailed description of the illustrative embodiment exemplifying the best mode of carrying out the invention as presently perceived.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] The detailed description of the drawings particularly refers to the accompanying figures in which:

[0009] FIG. **1** shows a perspective view of a linear shaped charge support.

[0010] FIG. **2** shows an overhead view of a linear shaped charge support.

[0011] FIG. **3** shows a side view of a linear shaped charge support.

[0012] FIG. **4** shows a perspective view of a linear shaped charge support supporting a linear shaped charge in a container.

[0013] FIG. **5** shows a perspective view of a linear shaped charge support supporting a linear shaped charge in a container.

[0014] FIG. **6** shows an overhead view of a linear shaped charge support supporting a linear shaped charge in a container.

DETAILED DESCRIPTION OF THE DRAWINGS

[0015] The embodiments of the invention described herein are not intended to be exhaustive or to limit the invention to precise forms disclosed. Rather, the embodiments selected for description have been chosen to enable one skilled in the art to practice the invention.

[0016] Generally, provided is a holder for stabilizing a linear shaped charge during epoxy curing comprising: a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from the first and the second end; and a pair of curved legs extending from the stair-step risers; wherein the pair of curved legs lock the support onto a container; wherein a linear shaped charge is positioned within the central aperture and extends downward into the container where it is stabilized while a curing agent within the container hardens.

[0017] In an illustrative embodiment, the linear shaped charge comprises a chevron or a V-shaped cross-section. In an illustrative embodiment, the container comprises a lip, an open top, a bottom, sidewalls, and a round cross-section. In an illustrative embodiment, the central aperture is chevron or a V-shaped. In an illustrative embodiment, the stair-step risers vertically offset the rectangular midsection from the container to prevent wicking of the curing agent up the linear shaped charge.

[0018] In an illustrative embodiment, provided is a holder for stabilizing a linear shaped charge during epoxy curing comprising: a container comprising; a lip, an open top, a bottom, sidewalls, and a round cross-section; and a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from the first and the second end; and a pair of curved legs extending from the stair-step risers; wherein the pair of curved legs match the round cross-section to permit locking of the support onto the lip of the container; wherein a linear shaped charge is positioned within the central aperture and extends downward into the container where it is stabilized while a curing agent within the container hardens.

[0019] In an illustrative embodiment, the linear shaped charge comprises a chevron or a V-shaped cross-section. In an illustrative embodiment, the container comprises a lip, an open top, a bottom, sidewalls, and a round cross-section. In an illustrative embodiment, the central aperture is chevron or a V-shaped. In an illustrative embodiment, the stair-step risers vertically offset the rectangular midsection from the container to prevent wicking of the curing agent up the linear shaped charge.

[0020] In an illustrative embodiment, provided is a system for stabilizing a linear shaped charge during epoxy curing comprising: a linear shaped charge; a container comprising; a lip, an open top, a bottom, sidewalls, and a round cross-section; and a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from the first and the second end; and a pair of curved legs extending from the stair-step risers; wherein the pair of curved legs match the round cross-section to permit locking of the support onto the lip of the container; wherein the linear shaped charge is positioned within the central aperture and extends downward into the container where it is stabilized while a curing agent within the container hardens.

[0021] In an illustrative embodiment, the linear shaped charge comprises a chevron or a V-shaped cross-section. In an illustrative embodiment, the central aperture is chevron or a V-shaped. In an illustrative embodiment, the stair-step risers vertically offset the rectangular midsection from the container to prevent wicking of the curing agent up the linear shaped charge.

[0022] FIG. **1** shows a perspective view of a linear shaped charge support **101**, FIG. **2** shows an overhead view of a linear shaped charge support **101**, FIG. **3** shows a side view of a linear shaped charge support **101**. In an illustrative embodiment, the linear shaped charge support **101** comprises a rectangular midsection **102** with a first and a second end **103**, **104**, and a central aperture **105**; a pair of stair-step risers **106**, **107** extending down from the first and second end **103**, **104**; and a pair of curved legs **108**, **109** extending from the stair-step risers **106**, **107**. In an illustrative embodiment, the pair of curved legs **108**, **109** lock the support **101** onto a container (shown below). In an illustrative embodiment, a linear shaped charge (shown below) is positioned within the central aperture **105** and extends downward into the container where it is stabilized while a curing agent within the container hardens. In an illustrative embodiment, the central aperture **105** can be sized and shaped to accommodate a linear shaped charge. In an illustrative embodiment, the central aperture **105** can be chevron or a V-shaped.

[0023] FIG. **4** shows a perspective view of a linear shaped charge support **101** supporting a linear shaped charge **401** in a container **402**. In an illustrative embodiment, the linear shaped charge **401** is a conventional linear shaped charge with a continuous explosive core enclosed in a metal lining. In an illustrative embodiment, the linear shaped charge **401** comprises a chevron or a V-shaped cross-section.

[0024] FIG. **5** shows a perspective view of a linear shaped charge support supporting a linear shaped charge **401** in a container **402**. In an illustrative embodiment, the pot (container) **402** comprises a lip **501**, an open top **502**, a bottom **503**, sidewalls **504**, and a round cross-section **505**. In an illustrative embodiment, the stair-step risers **106**, **107** vertically offset the rectangular midsection **102** of the support **101** from the lip **501** of the container **402** to prevent wicking of the curing agent (in some embodiments, epoxy) up the sides of the linear shaped charge **401**, which can otherwise cause the components (the linear shaped charge **401** and support **101**) to fuse together. Testing showed that a non-raised midsection provides the opportunity for significant interaction (wicking of epoxy) between the support **101**, the curing agent being used and the linear shaped charge **401**. At the center of the support **101** is a rectangular midsection **102** comprising central aperture **105** that is used to stabilize the linear shaped charge **401** during curing. The is linear shaped charge **401** is positioned vertically through the central aperture **105** for curing.

[0025] FIG. **6** shows an overhead view of a linear shaped charge support **101** supporting a linear shaped charge **401** in a container **402**. In an illustrative embodiment, the pair of curved legs **108**, **109** match the round cross-section **505** of the container **402** to permit locking of the support **101** onto the lip **501** of the container **402**. In an illustrative embodiment, the container **402** is a small metallurgical pot used for munitions analysis. As can be appreciated, the locking of the curved legs **108**, **109** keep the support **101** from moving up and down, and the curved nature of the curved legs **108**, **109** prevents the support **101** from changing its orientation aside from rotating about the center axis of the container **402**.

[0026] To use the device, a support **101** is locked in place on a container **402**, and a linear shaped charge **401** is inserted through the central aperture **105** into the container **402** containing a curing medium. The linear shaped charge **401** is placed as vertically as possible, and vertical verification is performed roughly **90** minutes after setting up the process to ensure proper orientation of the linear shaped charge **401** during curing. The timing for the check may vary slightly depending on environmental effects to curing time, such as humidity or temperature. After the material has cured, the support **101** is removed from the top of the container **402** and kept for future use.

[0027] Use of the inventive support compared to previous methods removes the need for one end of the linear shaped charge to be sanded, saving more than three hours of labor per linear shaped charge lot. Additionally, the curing epoxy needs to be checked only once, which is far less than the multitude of checks required without the support. The support is designed not to bond to the epoxy during the curing process, thereby allowing it to be reused.
[0028] In an illustrative embodiment, the support can be used to hold a variety of items for vertical metallurgical potting in different substances, through the change of the geometry of the central aperture. In an illustrative embodiment, the inventive support can be used to hold a linear shaped charge upright in acid to etch a sheath for grain structure analysis.
[0029] Although the invention has been described in detail with reference to certain preferred embodiments, variations and modifications exist within the spirit and scope of the invention as described and defined in the following claims.

Claims

1. A holder for stabilizing a linear shaped charge during epoxy curing comprising: a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from said first and said second end; and a pair of curved legs extending from said stair-step risers; wherein said pair of curved legs lock said support onto a container; wherein a linear shaped charge is positioned within said central aperture and extends downward into said container where it is stabilized while a curing agent within said container hardens.
2. The device of claim 1, wherein said linear shaped charge comprises a chevron or a V-shaped cross-section.
3. The device of claim 1, wherein said container comprises a lip, an open top, a bottom, sidewalls, and a round cross-section.
4. The device of claim 1, wherein said central aperture is chevron or a V-shaped.
5. The device of claim 1, wherein said stair-step risers vertically offset said rectangular midsection from said container to prevent wicking of said curing agent up said linear shaped charge.
6. A holder for stabilizing a linear shaped charge during epoxy curing comprising: a container comprising; a lip, an open top, a bottom, sidewalls, and a round cross-section; and a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from said first and said second end; and a pair of curved legs extending from said stair-step risers; wherein said pair of curved legs match said round cross-section to permit locking of said support onto said lip of said container; wherein a linear shaped charge is positioned within said central aperture and extends downward into said container where it is stabilized while a curing agent within said container hardens.
7. The device of claim 5, wherein said linear shaped charge comprises a chevron or a V-shaped cross-section.
8. The device of claim 5, wherein said container comprises a lip, an open top, a bottom, sidewalls, and a round cross-section.
9. The device of claim 5, wherein said central aperture is chevron or a V-shaped.
10. The device of claim 5, wherein said stair-step risers vertically offset said rectangular midsection from said container to prevent wicking of said curing agent up said linear shaped charge.
11. A system for stabilizing a linear shaped charge during epoxy curing comprising: a linear shaped charge; a container comprising; a lip, an open top, a bottom, sidewalls, and a round cross-section; and a support comprising; a rectangular midsection with a first and a second end, and a central aperture; a pair of stair-step risers extending down from said first and said second end; and a pair of curved legs extending from said stair-step risers; wherein said pair of curved legs match said round cross-section to permit locking of said support onto said lip of said container; wherein said linear

shaped charge is positioned within said central aperture and extends downward into said container where it is stabilized while a curing agent within said container hardens.

12. The device of claim 10, wherein said linear shaped charge comprises a chevron or a V-shaped cross-section.

13. The device of claim 10, wherein said central aperture is chevron or a V-shaped.

14. The device of claim 10, wherein said stair-step risers vertically offset said rectangular midsection from said container to prevent wicking of said curing agent up said linear shaped charge.